MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

STUDY GUIDE COPUOS

Agenda 1 : Discussing Sustainable Utilisation of Space Resources

Agenda 2 : Reviewing Implementation of Space2030 Agenda

Letter from R&D

Dear Delegates,

The Department of Research & Development of MUNSociety MPSTME welcomes you to the 11th edition of Mumbai MUN! Our department upholds a proud tradition of facilitating meaningful discourse on national and international matters, encompassing both historical events and current issues.

We are primarily tasked with formulating agendas and preparing comprehensive study guides. We have created detailed and engaging guides aimed at providing valuable resources for all delegates.

This year's theme, "Redefine the Norm," challenges us to think beyond conventional solutions and embrace innovative approaches to global challenges. In a rapidly evolving world, traditional methods often fall short of addressing contemporary issues. We believe that transformative change requires the courage to question established practices and the creativity to envision alternative pathways.

With this theme in mind, we have thoroughly picked agendas and committees that demand fresh perspectives, challenge conventional thought, and encourage innovative approaches to problem-solving. In this study guide, you will find well-researched background information to help you prepare for your committee sessions.

We, as facilitators, aim to provide you with the intellectual arsenal to combat any challenges that you might face and make the most of your Mumbai MUN experience.

We look forward to your active participation and contributions during Mumbai MUN and are confident that, together, we can make this conference a memorable and enriching experience for all.

Best wishes,
Department of Research & Development,
MUNSociety MPSTME.

Contents

Note: The QARMAs will be released pre-conference, post consultation with the Executive Board.

Agenda 1: Discussing Sustainable Utilisation of Space Resources

COMMITTEE OVERVIEW

The General Assembly established the Committee on the Peaceful Uses of Outer Space (COPUOS) in 1959 to oversee the exploration and utilisation of outer space and ensure it serves all of humanity by promoting peace, security and development. The Committee has two subsidiary bodies namely the Scientific and Technical Subcommittee and the Legal Subcommittee, both established in 1961. COPUOS was central in drafting the five treaties and five principles that guide outer space activities.[1] Each year, it serves as a forum for discussions on international cooperation in space exploration and the use of space technology to advance global development goals. With the rapid pace of advancements in space technology, the Committee acts as a critical platform for monitoring and discussing emerging trends and the evolution of the space agenda.

MANDATE

The committee's responsibilities include assessing peaceful space cooperation, exploring potential space initiatives that can be undertaken by the United Nations (UN), promoting space research programmes, and addressing legal issues related to space exploration. Its overall mandate, along with that of its two Subcommittees, focuses on strengthening the international legal framework governing outer space to facilitate global collaboration. Additionally, COPUOS supports efforts at the national, regional, and worldwide levels, including those of UN bodies and space organisations, to maximise the benefits of space technology and science. The overall goal is to improve global cooperation and efficiency in space activities.[2]

AGENDA OVERVIEW

As humanity's interest in space exploration grows, the sustainable utilisation of space resources has become a critical issue. With recent advancements in technology and exploration missions, there is increasing potential to extract and use resources from celestial bodies such as the Moon, Mars, and asteroids. Essential resources like water, rare metals, and minerals could support the construction of future habitats, fuel missions, and enable long-term human presence in space.

This agenda explores the challenges and ethical considerations of space resource utilisation, focusing on the balance between scientific exploration and environmental stewardship. Discussions will revolve around frameworks for international cooperation, strategies to prevent the monopolisation of resources, and policies for managing space debris to maintain the ecological integrity of celestial bodies. Establishing guidelines now will allow the global community to approach space exploration responsibly, honouring both the spirit of scientific discovery and the need for sustainability.

Key issues for discussion include developing international legal frameworks for resource utilisation and preserving the natural state of celestial bodies. By encouraging collaboration and innovation in sustainable extraction technologies, this agenda seeks to create

[1] *COPUOS*. (n.d.). UNOOSA https://www.unoosa.org/oosa/en/ourwork/copuos/index.html

[2] *COPUOS: Committee and Subcommittees*. (n.d.). UNOOSA. https://www.unoosa.org/oosa/de/ourwork/copuos/comm-subcomms.html

actionable policies that ensure space remains a shared resource for all of humanity's future.

THREATS

Exponential Growth in Space Objects and Actors

In current times, the avenue of space has been seeing a growth in the participation of actors like countries, academia, and the private sector. Asia has been at the forefront of this increased involvement with countries like India, China, Japan and South Korea increasing their national space programs and supporting commercial space sectors with investment incentives. For instance:[3]

India aims to perform 30 launches between 2024 and 2025. China on the other hand was responsible for one-third of the 180 successful space launches worldwide in 2022 and plans to conduct 100 such missions in 2024. Additionally, in May 2023, South Korea became the seventh country to launch a domestically developed rocket and launch vehicle. This underscores the growing interest various nations are taking in space exploration.

Moreover, one of the main reasons for the introduction of new actors in space is because of the commercialisation of space. It is no longer the exclusive domain of national governments. SpaceX (U.S.), Galactic Energy (China), MDA (Canada), and Skyroot (India) are some examples of private-sector businesses that are developing their independent launch capabilities. This increased involvement of the private sector is a result of the lowered barriers to entry in space exploration. Improved technologies, mass production of space objects and downsizing of satellites have enabled this. Innovation has made developing new space systems and launching payloads more affordable, allowing a broader range of organisations to engage in the space sector.

This increase in actors involved in space exploration in turn has led to an exponential increase in space objects. Space objects here are defined as crewed spacecraft, satellites, landers, probes and space station flight elements launched into Earth orbit or beyond.[4] According to data from the United Nations Office for Outer Space Affairs (UNOOSA),[5] 2,664 objects were launched into space in 2023, which was greater than any year before. Additionally, based on proposed space projects backed by various private and government organisations, as many as 12,000 new satellites could be launched into space each year over the next decade.[6] A peak in satellite concentration in low-Earth orbit has been observed with two-thirds of all active satellites (over 6,000) currently situated at altitudes ranging from 500 to 600 km. In low-Earth orbits, the frequency of events prompting collision avoidance procedures is on the rise, due

[3]Nadarajah, H. (2024, April 10). *The Emergence of New Actors in Space*. Asia Pacific Foundation of Canada. https://www.asiapacific.ca/publication/space-intro-emergence-new-actors

[4]*Annual number of objects launched into space*. (2024, January 4). Our World in Data. https://ourworldindata.org/grapher/yearly-number-of-objects-launched-into-outer-space

[5]*Online Index of Objects Launched into Outer Space*. (n.d.). UNOOSA. https://www.unoosa.org/oosa/osoindex/search-ng.jspx

[6]Harrington, J. (2022, March 29). *The future space industry: McKinsey aerospace experts look to 2030 | McKinsey*. McKinsey & Company. https://www.mckinsey.com/industries/aerospace-and-defense/our-insights/outer-space-in-2030

in part to increasing traffic congestion and the growing amount of debris.[7]

Space Debris and Risk of Collisions

Space debris, or orbital debris is any non-functional piece of machinery left in space by humans.[8] It includes satellites that are no longer functional, spent upper stages, paint flecks, astronaut detritus (which are items left behind by astronauts like cables, tools etc) and mission-related objects (such as launch adapters and lens covers).[9] Today, there are around 26000 pieces of debris in space that are larger than 10 cm in size. Additionally, the number of orbital debris that are larger than 1 cm is over a million.[10]

The reason space debris is a matter of concern is because of its potential to cause collisions. Most space debris travels at a very high speed of 18,000 miles per hour, which is almost seven times faster than a bullet.[11] This turns hazardous as collisions of such fast-moving debris can be catastrophic. Each piece of debris makes it increasingly difficult for functional machinery to avoid collisions.[12] This debris also affects the International Space Station (ISS) which has to conduct manoeuvring operations each year to avoid collision with such space junk.

Moreover, by 2030, more than 100,000 new spacecraft are estimated to be launched, in addition to the 8000 spacecraft currently in space. This could lead to overcrowding of space, further increasing the risk of collisions. This could potentially induce a self-sustaining chain reaction because each collision creates millions of pieces of shrapnel that are individually very hazardous. This is known as the "Kessler Syndrome", proposed by NASA scientist Donald J. Kessler in 1978.[13]

Space Militarisation

Space militarisation poses a threat as countries increasingly develop space-based communications, intelligence capabilities, and anti-satellite weaponry, raising the risk of a contested domain for military supremacy. The international community is divided between those seeking legally binding agreements to prevent an arms race and others advocating for voluntary norms and transparency measures. This highlights concerns that space could become

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[7] *ESA - ESA Space Environment Report 2024*. (2024, July 19). European Space Agency. https://www.esa.int/Space_Safety/Space_Debris/ESA_Space_Environment_Report_2024

[8] *What is space junk and why is it a problem?* (n.d.). National History Museum. https://www.nhm.ac.uk/discover/what-is-space-junk-and-why-is-it-a-problem.html

[9] *ESA - About space debris*. (n.d.). European Space Agency. https://www.esa.int/Space_Safety/Space_Debris/About_space_debris

[10] Kessler, D. (n.d.). *ESA'S ANNUAL SPACE ENVIRONMENT REPORT*. Space Debris User Portal. https://www.sdo.esoc.esa.int/environment_report/Space_Environment_Report_latest.pdf

[11] *Space Debris*. (n.d.). NASA. https://www.nasa.gov/headquarters/library/find/bibliographies/space-debris/

[12] *Space debris*. (2023, October 25). Interconnected Disaster Risks. https://interconnectedrisks.org/tipping-points/space-debris

[13] *Tackling the growing risks of space debris in Earth orbit – UK Space Agency blog*. (2023, November 6). UK Space Agency blog. https://space.blog.gov.uk/2023/11/06/tackling-the-growing-risks-of-space-debris-in-earth-orbit/

a new battlefield without cohesive regulatory frameworks.[14]

Moreover, with the increased number of space actors, there is a surge in the use of satellites for military surveillance.[15] The possibility of countries monitoring each other's activities from space has significant implications for national security and privacy. This could lead to tensions and distrust among nations.

Additionally, control over strategic space resources (such as water on the Moon or rare minerals on asteroids) could become a focal point for international rivalries. Countries may seek to claim and defend specific regions, undermining global stability and sparking conflicts over resource ownership.

Launch of the U.S. Space Force
Source: U.S. Space Force

CHALLENGES

Lack of a Comprehensive Policy and Regulatory Framework

One of the major challenges in tackling the problem of sustainable use of space resources is the lack of a robust regulatory framework on a global scale. The few conventions and agreements that exist are:[16]

The UN Outer Space Treaty of 1967 requires signatories to use outer space for peaceful purposes and prohibits the placement of nuclear weapons in space.

The Liability Convention of 1972 holds signatories liable for damages caused by their space objects to other states.

The Registration Convention of 1974, mandates signatories to register space objects with the UN and provide details about their orbits and purpose.

While these agreements cover key aspects of space debris, they don't address the need to reduce debris creation. Moreover, implementing these agreements is challenging because the origins of many objects are unknown. Space Debris Mitigation Guidelines (2007)[17] and the recently signed Artemis Accords specifically mention orbital debris, requiring that signatories take measures to minimise space debris and ensure the long-term sustainability of outer

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[14]'Outer space should never be an arena for militarization', delegate tells General Assembly debate on Moscow's veto of resolution aimed at curbing arms race | Meetings coverage and press releases. (2024, May 6). https://press.un.org/en/2024/ga12597.doc.htm

[15]Wehtje, B. (2023, December 19). Increased militarisation of space – a new realm of security. Beyond the Horizon ISSG. https://behorizon.org/increased-militarisation-of-space-a-new-realm-of-security/

[16]Kessler, D. (n.d.). *Technical Report: Space debris. Interconnected Disaster Risks 2023: Risk Tipping Points*. Amazon S3. https://s3.eu-central-1.amazonaws.com/interconnectedrisks/reports/2023/TR_231115_Space_Debris.pdf

[17]*OFFICE FOR OUTER SPACE AFFAIRS - Space Debris Mitigation Guidelines of the Committee on the Peaceful Uses of Outer Space*. (n.d.). UNOOSA. https://www.unoosa.org/pdf/publications/st_space_49E.pdf

space activities.[18] This includes committing to practices such as debris mitigation and safe end-of-life disposal for space objects. As of June 2023, 27 states have signed the Artemis accords; however, they are explicitly a "non-binding set of principles" rather than enforceable regulations.

While some countries have established national regulations, there are currently no binding international rules for space debris mitigation. UNOOSA, IADC, and various space agencies and governments have created guidelines and best practices to reduce debris accumulation and stabilise the orbital environment. These measures include avoiding intentional debris generation (such as anti-satellite tests), preventing accidental explosions in orbit, conducting collision avoidance when feasible, and recommending the deorbiting of satellites 25 years after their operational life ends. However, these are merely guidelines and adherence to these measures is lacking.[19]

Only 60% of operators in low Earth orbit follow the guidelines, with compliance dropping to less than 20 per cent in orbits above 650 kilometres. The ESA's Annual Space Environment report indicates that while compliance with space debris mitigation practices is slowly improving, the current level of implementation is still inadequate to ensure long-term sustainability in the environment.[20]

Lack of International Cooperation

While national regulations have fostered innovation, the absence of a unified international framework creates significant challenges. Article 2 of the Outer Space Treaty (OST) prohibits national appropriation of outer space, complicating efforts to establish ownership rights over space resources and potentially leading to disputes among nations and private entities.

Moreover, Article 3 links space activities to international environmental law, emphasising the need for a legal framework to address environmental risks from resource extraction. Moreover, as of 2024, only 17 countries have ratified the Moon Agreement. This reflects a lack of unified consensus amongst nations regarding a common framework for sustainable space resource utilisation.

A comprehensive agreement on space resources is essential to ensure equitable access and benefit for all nations, not just the technologically advanced. Establishing such a framework would promote international cooperation, equitable resource sharing, and the preservation of outer space's environment.

Obstacles in Asteroid Mining

Asteroid mining is a proposed approach for extracting raw materials from asteroids and other minor celestial bodies. While still theoretical, it offers a potential solution to the increasing carbon pollution and resource depletion issues facing Earth.[21] However, there are various challenges associated with asteroid mining such as:

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[18] *Artemis Accords*. (n.d.). NASA. https://www.nasa.gov/artemis-accords/

[19] Kessler, D. (n.d.). *echnical Report: Space debris. Interconnected Disaster Risks 2023: Risk Tipping Points*. Amazon S3. https://s3.eu-central-1.amazonaws.com/interconnectedrisks/reports/2023/TR_231115_Space_Debris.pdf

[20] Kessler, D. (n.d.). *ESA'S ANNUAL SPACE ENVIRONMENT REPORT*. Space Debris User Portal. https://www.sdo.esoc.esa.int/environment_report/Space_Environment_Report_latest.pdf

[21] *What Is Asteroid Mining?* (n.d.). EVONA. https://www.evona.com/blog/what-is-asteroid-mining/

Economic viability

The costs of developing mining technologies and conducting operations in space are currently very high. Reducing these expenses is crucial in order to make asteroid mining a feasible endeavour in the future.[22]

Identifying asteroids suitable for mining

Not all asteroids contain valuable resources or are accessible for extraction. Hence, asteroid mining requires extensive surveying and analysis which makes it a time-consuming and resource-intensive process.

Conflicts between nations

Conflicting national interests and territorial claims can become a source of disputes between nations. This geopolitical tension can hinder collaboration and investment, ultimately delaying the progress of mining initiatives and the sustainable utilisation of space resources.

Inadequate Scientific and Technical Progress and ISRU

Inadequate scientific and technical progress poses a significant barrier to the sustainable utilisation of space resources. While interest in extracting materials from celestial bodies like asteroids and the Moon is growing, the technologies needed for efficient and safe resource extraction remain underdeveloped. NASA's Artemis program aims to establish a sustainable human presence on the Moon. However, the technologies for in-situ resource utilisation (ISRU)[23]—essential for converting lunar materials into usable resources—are still largely experimental. Advancements in ISRU technologies are crucial for reducing reliance on Earth-supplied resources, yet current research and development efforts are insufficient to meet projected timelines for lunar missions.

Additionally, the environmental impacts of space resource extraction are not well understood, complicating sustainability efforts. Research shows that mining extraterrestrial bodies could lead to unforeseen ecological consequences, such as contamination of celestial environments. Without adequate scientific understanding and technical readiness, efforts to sustainably utilise space resources risk causing irreversible damage. This could undermine the long-term viability of space exploration. This situation highlights the urgent need for increased investment in research and collaboration. Space agencies, academia, and industry must work together to develop the technologies and knowledge required for sustainable space resource management.

Safety Risks in Space Operations

Space Operations never tend to seem easy as there is always a safety risk linked to it. There are five main hazards listed by NASA—space radiation, isolation, lack of gravity, distance from Earth and lastly, closed environments. The most threatening out of the bunch is space radiation as it can cause serious harm to the human body. But it is also necessary to note that the rest can also aggravate the damage done to the human body and mind. To overcome the impacts of hostility and confined environments, one can only learn to adapt to the situation and even to obtain new resources from Earth becomes more challenging than usual. This poses a serious threat and risk to the person travelling in space.

There have been targets set to resolve these problems with the help of vigorous research done in the laboratory and primarily in the International Space Station. In order to

[22]Prete, R. (2024, July 15). *Asteroid Mining: Unveiling the Economic and Technological Frontier*. Linqto. https://www.linqto.com/blog/asteroid-mining-challenges-and-potentials/

[23]*In-Situ Resource Utilization (ISRU)*. (n.d.). NASA. https://www.nasa.gov/mission/in-situ-resource-utilization-isru

extrapolate to multi-year interplanetary missions, NASA is properly understanding the cores of the human minds and the reactions or stimulations felt during the extended forays into space. The data derived from this plays a huge role in developing methodologies in order to achieve high targets.[24]

Resource Depletion and Mining

Resource depletion in space, particularly regarding the mining of celestial bodies, poses significant sustainability challenges. As interest in mining asteroids and the Moon grows, there are concerns about the environmental impact and potential conflicts arising from resource extraction. The unregulated nature of these activities could lead to the over-exploitation of space resources, threatening the long-term sustainability of outer space environments.

CASE STUDY

India

The Space Program of India

ISRO (Indian Space Research Organisation)

Established in 1969, ISRO seems to be a prominent figure in Space Research & development for the peaceful usage of space resources.[25]

Space Vision 2030

The long-term vision for India, focuses on overcoming societal challenges and development requirements through the use of space technology with optimum resource management and sustainability.

A Paradigm Shift in India's Space Policy

Draft Space Policy 2020

The current draft policy of India offers encouragement to the private sector to engage in outer space activities. The policy attempts to achieve any commercial use of space activities and resources with a regulated sustainable approach in compliance with international laws including treaties such as the Outer Space Treaty of 1967.

IN-SPACe

The Constitution of the Indian National Space Promotion and Authorization Centre (IN-SPACe), established in 2020, promotes and authorises private sector activities while ensuring the responsible and sustainable development of space resources.

India's Endeavour to Meet the Principles set forth under COPUOS

India being one of the oldest member countries of COPUOS endeavours to follow its goals and objectives for defence and security purposes in an environment-friendly manner.

International Cooperation

Outside India, space debris management, regulations surrounding space resources mining, and outer space resource long-term sustainability have involved Indian negotiations through the COPUOS framework.

Chandrayaan Missions

Chandrayaan-1

This enabled India to be one of the first countries to successfully map and target water-bearing

[24] 5 *Hazards of Human Spaceflight*. (n.d.). NASA. https://www.nasa.gov/hrp/hazards/

[25] Encyclopædia Britannica, inc. (2024b, October 24). *Indian Space Research Organisation*. Encyclopædia Britannica. https://www.britannica.com/topic/Indian-Space-Research-Organisation

regions of the Moon and therefore laid the groundwork for future attempts to gather lunar resources.[26]

Chandrayaan-2

With this mission, India developed its capabilities further to examine the possible opportunities in the South Pole region of the moon which is important for future settlements and mining as it is considered to be an ice-deposit-rich region.

Chandrayaan-3 (2023)

This consistent further exploration followed by detailed studies of the lunar surface helps to build a foundation for the use of lunar resources for ISRU missions.

Pre Launch LVM3-V4 of Chandrayaan 3
Source: ISRO

Role in Space Debris Mitigation

Netra Project (Network for Space Object Tracking and Analysis)

It is a program that aims to track and analyse space debris and offers an independent Indian perspective regarding debris management. This initiative forms part of a larger plan in India, which seeks to manage debris threats and prevent the pollution of resources to promote the sustainable use of space.

Adopt a Sustainable Approach

Even space missions envisaged by India such as Mars and Asteroid mining, will take into account sustainability as of now and in future: environmental as well as economics.

India's Approach to Exploring Space Resources Sustainably

India's strategy includes the development of Green Satellites for sustainable space exploration, promoting prudent utilisation of space resources, and application of solar energy-based ISRU technologies to minimise terrestrial resource usage.

Cosmically, India intends to be a major global contributor towards peaceful space resources exploration and their sustainable use to leverage the international space economy, whilst abiding by COPUOS principles.

The perspective of India's space program is compatible with the peaceful use of outer space resources. From ISRO's efficient missions and participation in the United Nations platforms including COPUOS, India's concern for the sustainable development of outer space resources is evident and demonstrates a commitment to international cooperation,

[26]NASA. (n.d.). *Chandrayaan-1 / moon impact probe - NASA science*. NASA. https://science.nasa.gov/mission/chandrayaan-1/

technological progress, and environmental sustainability.

PAST INTERNATIONAL ACTIONS

World Economic Forum

Framework for Sustainable Development

The WEF addresses how space technology can be key in the achievement of the United Nations' Sustainable Development Goals. For example, satellite data is important in the tracking of global climate change (SDG 13), improvement of agricultural practices (SDG 2), and increasing the resilience of any disaster-prone area (SDG 11). The Forum initiated global challenges in leveraging space data and unlocking research and development for health, environment, and governance. The WEF sees PPPs as instrumental in achieving the vision of advancing space technology through collaboration among private companies and their interaction with government, academia, and various other teams toward innovation in sustainable space practices. The Forum has been tasked with focusing public attention on the need for sustainable practices in humanity's exploration and utilisation of space resources.[27]

The United Nations Space Debris Mitigation Guidelines (2007)

The United Nations Space Debris Mitigation Guidelines, developed by the Committee on the Peaceful Uses of Outer Space (COPUOS) and endorsed by the General Assembly in 2007, outline strategies for minimising the creation and impact of space debris. These voluntary guidelines focus on sustainable practices for both new and existing spacecraft operations. Key recommendations include limiting debris release during standard operations, preventing break-ups by addressing failure risks, minimising accidental collision probabilities, and avoiding intentional on-orbit destruction. They also advise passivating or removing spacecraft from low-Earth orbit after missions to prevent long-term debris presence.[28]

The guidelines are informed by existing standards and best practices, such as those from the Inter-Agency Space Debris Coordination Committee (IADC), and aim to improve the safety of space operations. Although they are not legally binding, UN Member States are encouraged to adopt these practices to mitigate risks to current and future space missions through national regulations or similar mechanisms.

Long-term Sustainability Guidelines for Outer Space Activities (2019)

The Long-term Sustainability Guidelines for Outer Space Activities, adopted by the United Nations Committee on the Peaceful Uses of Outer Space (UN COPUOS) in June 2019, provide a framework aimed at ensuring safe and responsible practices in outer space. Comprising 21 voluntary, non-binding guidelines, these recommendations are designed to assist states and other stakeholders in developing their space activities while safeguarding the outer space environment.

The guidelines emphasise the importance of cooperation among nations, encouraging improved coordination and information exchange among all space actors, including governmental, private sector, and civil society entities. They address every phase of space missions—from pre-launch planning to

[27]*The path forward for Sustainable Deep Space Exploration*. World Economic Forum. (n.d.). https://www.weforum.org/agenda/2024/07/sustainable-space-exploration-path-forward/

[28]*OFFICE FOR OUTER SPACE AFFAIRS - Space Debris Mitigation Guidelines of the Committee on the Peaceful Uses of Outer Space*. (n.d.). UNOOSA. https://www.unoosa.org/pdf/publications/st_space_49E.pdf

operational activities and end-of-life disposal—highlighting the need for comprehensive strategies that prevent harm to both space infrastructure and the environment.

CONCLUSION

In conclusion, the sustainable utilisation of space resources offers a transformative opportunity for humanity, but it also poses significant challenges, particularly regarding space debris. As we explore and extract materials from celestial bodies, we must implement responsible practices to prevent further accumulation of debris in Earth's orbit, which can jeopardise future missions. Moreover, international collaboration and cooperation between all actors is essential in order to innovate technologies that can help us harness the full potential of space.

Agenda 2: Reviewing Implementation of Space2030 Agenda

AGENDA OVERVIEW

The Space2030 Agenda serves as a visionary framework for the global space community, outlining essential guidelines for achieving the Sustainable Development Goals. With international cooperation at its core, this agenda aims to broaden horizons for transformative advancements in space technology and innovation. It is centred around its four main objectives - space economy, space society, space accessibility, and space diplomacy. Its first and foremost aim is to ensure the equitable distribution of space resources among all nations, especially to uplift developing countries and their space programs. To achieve the goals of the Space2030 Agenda, it is crucial to identify and address gaps in collaboration and integration with initiatives. A review of partnerships among member states and United Nations bodies is needed to enhance the development of space resources for global benefit, ensuring that advancements in space technology lead to meaningful improvements in life on Earth by 2030.

INTRODUCTION

The Space2030 agenda is a landmark agenda endorsing space as the driving force of sustainable development and charting a way to improve the contribution of space and its applications. It aims to align space goals with the sustainable development goals.[29] This agenda was the culmination of a three-year-long process containing regional and international mechanisms, programmes and projects that its member states can gain the advantage of. It offers a myriad of resources through which its member states can harness space technologies, applications and space-derived data to enhance their economic growth. It also wishes to provide a platform for global collaboration and cooperation towards space exploration and peaceful usage of outer space.

BACKGROUND

Goals

The Space2030 agenda includes four major objectives structured around space economy, space society, space accessibility and space diplomacy.[30] These pillars of the agenda are mutually enhancing and reinforcing. Expansion of the space economy by increasing access to space leads to overall growth of the space community while also promoting international cooperation. This way they form a cohesive network to support each other.

Enhance space-derived economic benefits and strengthen the role of the space sector as a major driver of sustainable development

This includes raising awareness about the significance of space science and technology. It aims to integrate the space industry along with the environment, health, and energy to provide unique solutions for the socio-economic development of its member states. It also includes the provisions to be able to provide equal space access for all keeping in mind all safety norms as well. It aims to develop the space industry focusing especially on small-sized enterprises and indirectly extending its benefits to non-space sectors as well.

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[29] *Committee on the Peaceful Uses of Outer Space endorses the "Space2030" Agenda and advances its agenda on several key items*. (2021, September 7). UNIS Vienna. https://unis.unvienna.org/unis/en/pressrels/2021/unisos557.html

[30] *A/RES/76/3 General Assembly*. (2021, October 28). UNOOSA. https://www.unoosa.org/res/oosadoc/data/resolutions/2021/general_assembly_76th_session/ares763_html/A_RES_76_3_E.pdf

Harness the potential of space to solve everyday challenges and leverage space-related innovation to improve the quality of life

The second objective aims to provide help and support to research focusing on space. The knowledge of applications of space technologies is extremely useful in fields of natural environment, climate, urbanisation and protection of all ecosystems. It emphasises on the fact that this can be used in the disaster management cycle, support global health and develop human settlements and infrastructure both urban as well as rural.

Improve access to space for all and ensure that all countries can benefit socioeconomically from space science and technology applications and space-based data, information and products, thereby supporting the achievement of the Sustainable Development Goals

It aims to inspire the young generations and ignite their interests towards space by supporting various initiatives for the youth. Specifically in developing countries, it aims to provide space education and training which can be used to enhance and promote space exploration beyond the Earth orbit and increase our knowledge of outer space to benefit humanity. It also aims to promote gender inclusivity including strengthening the role of women in space science, technology and engineering.

Build partnerships and strengthen international cooperation in the peaceful uses of outer space and in the global governance of outer space activities

The fourth and final objective is focused majorly on the implementation of this agenda with the cooperation of international entities. The United Nations Office of Outer Space Affairs (UNOOSA) plays a central role in this objective. This objective aims to develop international law related to outer space due to the emerging issues uncovered in recent times. It aims to strengthen technical assistance, enhance the safety of outer space endeavours and ensure the long-term sustainability of space activities. It promotes international collaboration towards the exchange of information and best practices. Moreover, it provides supervision to the space activities of non-governmental entities and ensures their consistency with the international laws set to facilitate the flourishing space industry.

Strategic Priorities

The United Nations recognises the importance of space science and technology in our everyday lives and how it provides a wide range of benefits to mankind. Space continues to be a source of motivation and creativity for humans through which the space community moves ahead with their pursuits. It emphasises how space tools are equally essential towards the fulfilment of the sustainable development goals, the Sendai Framework and the Paris Agreement where the "Space2030" agenda is a direct or indirect influence for it by providing indispensable information to monitor its progress. It highlights that under the international legal regime, the Outer Space Treaty forms its foundation to govern outer space ventures. It calls for the increased usage of unique technologies to ensure that the entire global community can reap its advantages. [31]

Tools and Resources

A wide range of mechanisms, projects and platforms are put in place or under development which contribute to the successful implementation of the Space2030 agenda. The first step of space development requires access to space-based data through which member states

[31] *Space2030 agenda: Space as a driver for peace*. (n.d.). UNOOSA. https://www.unoosa.org/oosa/en/outreach/events/2018/spacetrust.html

can carry out disaster reduction and emergency responses. This has been provided by the United Nations Platform for Space-based Information for Disaster Management and Emergency Response (UN-SPIDER) and the International Charter on Space and Major Disasters. This data is provided by a wide range of international agencies that focus on different aspects and collect necessary information. This includes the International Space Climate Observatory and the World Meteorological Organization Integrated Global Observing System (WIGOS) which execute climate monitoring and environmental activities. The UNOOSA also provides several other tools and initiatives as a method to provide equal access to space resources globally. Some of these tools include the Access to Space for All initiative and the Open Universe initiative to widen access for emerging spacefaring nations, the Space for Women project and Space for Youth to bring forth more young talent as well as provide women with equal prospects in the space industry. Finally, it urges countries with highly developed space programs to collaborate with other nations via the sharing of technology and infrastructure and provide additional resources for the implementation of this agenda.

Timeline

June to October 2018 - The member states of the international space community gathered to reflect on the achievements in space exploration and recognise its shortcomings leading to the formation of the Space2030 agenda.

2018 to 2021 - After the formation, the committee formulated an implementation plan and submitted it to the General Assembly for action.

October 2021 - The Space2030 agenda was adopted in the year 2021 at the 76th general assembly session. It aims for regular sessions to be undertaken amongst the member states to track the progress of the implementation of the agenda.

2025 - A midterm review will be held to assess the progress made by the member states and understand the complications faced in the process.

2030 - The final review will be held where the member states shall provide their results in front of the general assembly.

CHALLENGES

The Space2030 Agenda aims to utilise all potential from space science and technology to face global challenges and promote sustainable development. However, when this agenda is put into practice, there are a number of challenges it needs to consider. Hereafter, some of the key challenges related to regulatory and legal frameworks, long-term sustainability, and financing and resources are presented.

Regulatory and Legal Frameworks

International Treaties and Agreements: Such a legally defined framework as it stands today is derived mainly from international treaties created during the 1960s and 1970s of which the Outer Space Treaty (1967) and other related treaties are examples. The present legal frameworks do not provide for the required precision, as they do not usually correspond to very specific challenges like modern commercial space activities or space resource utilisation.

Regulations differ from country to country, sometimes creating confusion and hindering international cooperation.

Acquiring a licence for space missions is a complex process and varies with requirements in all countries. A lot of work has to be directed at coping with domestic regulations while remaining in line with international norms. This becomes difficult for private companies or emerging space-faring nations.

Space Traffic Management

One great concern would be space traffic management, especially with the increasing number of satellites in orbit. Effective frameworks are essential in order to coordinate launch and work with satellites to avoid collisions and control orbital debris.

With many commercial entities becoming involved with activities relevant to space, issues on ownership and protection of intellectual property rights related to space technology become increasingly relevant.

Liability for Damage

Liability for space damage, like collisions or debris, remains complex. The Outer Space Treaty and Liability Convention hold states accountable for damages from their space objects, but proving fault—especially with private actors involved—can be challenging. This uncertainty can hinder investment, leading to calls for clearer regulations to ensure accountability and safety in space.

Investment in R&D

Funding for R&D in sustainable technologies is low, whereas the future alone could meet investment needs. Governments and players in the private sector have to ensure that investments are prioritised when implementing the Space2030 Agenda.

Building effective public-private partnerships can be done through resource leveraging, but only if well-defined frameworks and the right incentives are in place to attract private investment.

The space industry is rapidly changing; however, market uncertainties often discourage investment. There should be clear policies and regulatory frameworks in place for a stable investment environment.

Sustainable operations in space may have increased front-end costs. The stakeholders need to find a way of striking the balance between sustainability and economic viability. International cooperation can enhance resource sharing and reduce costs. Building trust and effective mechanisms for cooperation among countries, however, remains a challenge.

The Space2030 Agenda promises a complex landscape of regulatory, environmental, and financial challenges. Therefore, this issue must be dealt with cooperatively among the interplay of governments, private sector stakeholders, and international organisations. Only then can the space community unlock the transformative potential for sustainable development by developing coherent regulatory frameworks, ensuring long-term sustainability and adequate financing.

PAST INTERNATIONAL ACTIONS

Outer Space Treaty (1967)

The Outer Space Treaty is a foundational landmark treatise providing a basic framework on international space law enforced in 1967. It establishes a structure outlining the methods by which member states shall conduct space activities and specifies regulations governing the utilisation of outer space.[32] One key principle is to ensure that space exploration is free for all states regardless of their developmental status and that it cannot be claimed by any nation by unlawful methods of occupation. The exploration of space and other celestial bodies should be used for enhancing knowledge and gaining informative benefits. This shall be done exclusively in peaceful ways. States shall be held accountable regarding the impacts of their space activities

[32] *The Outer Space Treaty*. (n.d.). UNOOSA. https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/introouterspacetreaty.html

and hence any endeavour which might contaminate outer space should be avoided at all costs. Regular meetings have been undertaken every year to track the progress of all member states where their expertise and challenges are shared.

Registration Convention (1976)

This treaty is essentially a registrar maintained by the United Nations containing all the objects launched into outer space. It was formed at the request of the member states in the Outer Space Treaty to help in the easy identification of all space objects launched by any other state. Using this, the states can bear complete legal accountability and liability for their space objects. After the formation of the Outer Space Treaty, many discussions were held after which the convention on registration of objects launched into outer space was adopted in 1976.[33] All states now have to maintain national registries and provide this information to the United States Secretary-General to include in the UN register. This has successfully led to more than 88% of all satellites, and other spacecraft being recognised and registered.

Moon Agreement (1979)

The Moon Agreement was implemented in 1979 recognizing the Moon's important role in outer space exploration. Hence keeping this in mind, this treaty aims to set forth ground rules for the use and exploration of the Moon. These rules also extend towards all other celestial bodies in space which may be explored in future times. It sets a base rule that no state has singular control over any part of the Moon. No means of violence, hostility, or military might shall be used on the moon against any other state. The Moon shall be used only for the benefit and interest of mankind and this requires the cooperation of all international entities who have the desired capabilities to achieve this. Any activities conducted on the Moon shall be informed to the Secretary-General to the greatest extent of information possible. This includes information regarding the purpose, location, equipment used, and scientific results gained to be provided at the earliest time possible after the completion of the mission. All member states are allowed to carry out any investigation they deem worthy. It also urged the states that any substances collected should be made available to other states as well in small sample quantities for better scientific investigation. All in all, the main aim is to provide equitable opportunities to all states without discrimination of any kind for advancements of their space programs.

Space Benefits Declaration (1996)

This treaty emphasises the topic of peaceful and beneficial uses of outer space. It has been established in previous treaties that space exploration should be for the advancement of knowledge of mankind. Special focus should be placed on providing assistance to developing countries to take these excursions. There should be increased efforts by countries with relevant capabilities to provide equitable distribution of information as well as guidance to others. This should be carried out in all modes which seem most feasible and helpful including global, multilateral and commercial methods of cooperation. It places importance on focusing on promoting the development of space technology in interested states by providing them with expertise from capable states. Finally, all countries are encouraged to contribute to other

[33] *United Nations Register of Objects Launched into Outer Space*. (n.d.). UNOOSA. https://www.unoosa.org/oosa/en/spaceobjectregister/index.html

United States programmes and initiatives in accordance with their capabilities. [34]

UNISPACE Conferences (1968, 1982, 1999, 2018)

Acknowledging the impact of space technologies on socioeconomic development, the United Nations introduced the initiative of the United Nations Conference on the Exploration and Peaceful Uses of Outer Space (UNISPACE) Conferences to emphasise the importance of collaboration and cooperation in outer space affairs. Facilitated by the UN, the three conferences primarily organised in the late 19th century helped in fabricating a global platform for states and international organisations to peacefully cooperate and have discussions over space exploration as well as considering the exploitation of the same. This yields tremendous benefits to the nations.

UNISPACE Conference
Source: UNOOSA

The first conference, held from 14 to 27 August 1968, mainly was to indicate the vast potential of space technologies and also raise awareness about the positive outcomes driven by it. The second conference, held from 9 to 21 August 1982, shed light on the violation of peace in the race to achieve independent glory. It focused on maintaining peaceful exploration and wise use of resources. There was also support provided to the developing countries in order to achieve crucial and essential space assets. The third conference, held from 9 to 30 July 1999, outlined the ways to protect the environment and ensure proper resource management while also creating an effective blueprint for the peaceful use of the resources. The recent conference held in July 2018 was to commemorate the fiftieth anniversary which was essentially held to ponder and assess the future of global space cooperation and its benefits to humankind.

CASE STUDY

International Space Station (ISS) Collaboration

The International Space Station is home to research breakthroughs and inventions that cannot be formulated on Earth. It was constructed and designed between 1984 and 1993 with the collaboration of NASA (United States), Roscosmos (Russia), ESA (Europe), JAXA (Japan), and CSA (Canada). It was first launched on 20th November 1998, currently marking its 25 years in orbit. There have been various expeditions hosted to learn more about space, the recent being Expedition 72.[35] The core mission of this expedition is to understand the space phenomena which can help in the transit of humans in and out of space and delve into the fields of pharmaceutical manufacturing, advanced life support systems, genetic sequencing in microgravity.

A gravitating shift will be marked by the extension of the International Space Station operations through 2030. With the hunger to explore more, for the past two decades, there has been a continuous human presence in the orbit around the Earth sent by the United Nations to

[34] *Space Benefits Declaration*. (n.d.). UNOOSA. https://www.unoosa.org/oosa/en/ourwork/spacelaw/principles/space-benefits-declaration.html

[35] *Expedition 72*. (2024, September 23). NASA. https://www.nasa.gov/mission/expedition-72/

corroborate and test the utilities of space technologies, conduct groundbreaking research and inculcate the necessary skills required to embark more further into deep space. The presence of over 4,200 researchers in the microgravity laboratory from across the world has resulted in more than 3,000 research investigations which are paying off as scientific, educational and technological benefits for the people on Earth. 110 countries have shown their enthusiasm and support by participating in the activities which involve 1,500,000 students per year in STEM activities.[36]

The spacecraft was crafted to last a long duration and NASA has ensured that the systems stay maintained and upgraded. But the technical lifetime is limited to the primary structure which in turn is dependent on the dynamic loading and orbital thermal cycling. Thus, NASA has planned to maintain its orbiting till the year 2030. The deorbiting of the spacecraft has played a major concern for which NASA and the ISS have come up with a variety of options to ensure a safe manoeuvre. One of the major risks affecting the populated areas, NASA announced SpaceX has been selected to develop and deliver the U.S. Deorbit Vehicle which will help in the easy deorbiting of the spacecraft.

Although there are some years left for the spacecraft to be in orbit, there are possibilities of the renewal sought by NASA to go past 2030. The prolonged work will bring out another productive decade of research advancement and by the late 2020s will open the doors to a seamless transition of the capabilities in low-Earth orbit to one or more commercially owned and operated destinations.

International Space Station
Source: NASA

UN-SPIDER

Established in 2008, the United Nations Platform for Space-based Information for Disaster Management and Emergency Response (UN-SPIDER) is an initiative implemented towards developing solutions that cater to address the disproportionate amount of specialised technologies available to the developing countries which can be effectively used in terms of managing disasters and narrow the risks linked with it. It works to understand and implement the core of ideologies of space-based information into a disaster management cycle which can help in overcoming various situations.

Using remote sensing, satellite-based telecommunication and global navigation satellite systems, has eased the processes of disaster risk management and handling emergency responses. The UN-SPIDER acts as a bridge in helping developing nations to utilise space-based resources so that they can brace all the phases of the disaster management cycle such as prevention and preparedness.. They have goals of sharing vast knowledge and effectively implementing space technologies in institutions. One crucial objective is to achieve a better flow of data and information based on disaster risks or impacts between all stakeholders and the residents affected by them.

[36]*ISS 2030: NASA Extends Operations of the International Space Station*. (2022, January 11). NASA Science. https://science.nasa.gov/resource/iss-2030-nasa-extends-operations-of-the-international-space-station/

Most importantly, they believe in catering to every country's needs in a personalised way, tailoring solutions or recommendations that suit the nation's conditions and also helping in training government staff. Through the conferences and workshops conducted, they impart the key ideas behind the relation of space and disaster management by involving the stakeholders of the same.[37]

KEY PARTIES INVOLVED

International Organizations

UN Office for Outer Space Affairs (UNOOSA)

UNOOSA is the lead UN entity tasked with the responsibility of promoting international cooperation for the peaceful use of outer space and ensuring that outer space activities are within the SDG frameworks.

The Office is involved in various activities, such as the Space2030 Agenda and the initiative on the Long-term Sustainability of Outer Space Activities, promoting best practices and guidelines on increasing the sustainability of space operations.

Committee on the Peaceful Uses of Outer Space (COPUOS)

This committee, which dates back to 1959, includes 95 member states and serves as an international forum for discussion related to space law and the peaceful use of outer space. COPUOS formulates international space laws, deals with issues related to space debris, and facilitates international cooperation in space science and technology.

International Telecommunication Union (ITU)

The ITU is a specialised agency of the UN, where matters concerning information and communication technologies are concerned, such as satellite orbit and frequencies. The coordination of satellite communications is of great importance since it is necessary to avoid interference and make possible the sustainable use of space resources.

World Meteorological Organization (WMO)

The WMO facilitates international cooperation in meteorology, climatology, and hydrology by taking advantage of space agency observations in climate monitoring and disaster response.It facilitates projects through the utilisation of satellite data in improving forecasting and resilience to climate change and hence entails a relation to the SDGs.

United Nations Educational, Scientific and Cultural Organization (UNESCO)

UNESCO promotes education and capacity building in areas associated with space science and technology; it will continue to foster opportunities for developing countries and international knowledge sharing.

Regional Space Programs

European Space Agency (ESA)

ESA is an intergovernmental organisation with a membership of 22 sovereign states, formed mainly to explore space; it plays an essential role in European capability development in space. Under Earth observation, ESA works on projects like Copernicus, which directly supports environmental monitoring and climate action.

African Space Agency (ASA)

ASA was established in the year 2018. The ASA was established to facilitate and promote space activities across the African continent. Projects on disaster management, agriculture, and resources

[37]*United Nations Platform for Space-based Information for Disaster Management and Emergency Response (UN-SPIDER)*. (n.d.). UNOOSA. https://www.unoosa.org/oosa/en/ourwork/un-spider/index.html

with the aim of capacity building in the regions for using space technology for sustainable development.

Indian Space Research Organisation (ISRO)

ISRO is known for its less costly space exploration missions and has offered many satellites for Earth observation, communication, and navigation. The Indian space program meets different developmental purposes by offering satellite data for agriculture, disaster management, and urban planning, which contributes to the overall Space2030 Agenda.

National Aeronautics and Space Administration (NASA)

As it is a national agency, NASA also works with many international organisations that advance research and activities involved with sustaining space. NASA's Earth Science Division uses satellite data to address climate change and natural disasters and meets the goals outlined in the Space2030 Agenda.

Regional Space Organizations

The ASCG (Arab Space Coordination Group) and the Asian Space Agency Consortium help advance cooperation between member states in different regional settings, enhancing the promotion of space technologies.

Private Bodies

SpaceX

Founded by Elon Musk, SpaceX has revolutionised space travel using its reusable rocket technology and low-cost launch services. It has contributed to the provision of affordable access to space, with the support of multiple missions such as satellite deployment for communication and Earth observation, which are supposed to contribute to achieving the Sustainable Development Goals.

Blue Origin

Part of Jeff Bezos's vision, Blue Origin works on technologies that will make private space travel and exploration a reality. The company aims to make a future where millions of people can be in space and live their lives, thereby focusing the company on sustainable practices in order to meet that vision.

Planet Labs

Planet Labs provides high-frequency Earth imaging for agriculture, forestry, and environmental monitoring using small satellites. Their data supports all sectors in making informed decisions relating to resource management and climate adaptation, hence directly contributing to the Space2030 Agenda.

Northrop Grumman

The company is engaged in space development, launch services, as well as space logistics for both Government and Commercial space missions. Northrop Grumman is actively engaged in some of the initiatives focused on sustainability in space activities, of which, are debris mitigation technologies.

New Start-ups

Several new start-ups are coming into the space industry with innovative answers in satellite technology, removal of space debris, and resource harvesting. Some examples of other companies moving forward with sustainable practices in the private space market include Astroscale, which specialises in cleaning up space debris, and Rocket Lab, which is involved in the launching of small satellites.

The successful implementation of the Space2030 Agenda leans on diverse coalitions of international bodies, regional space programs, and private entities. Their role is to fundamentally support each other in the practice of sustainable development in space and in cooperation with utilising technology for global development. With team effort, these stakeholders can address arising challenges in the efficient utilisation of space resources and contribute to the broader goals of sustainable development.

CONCLUSION

With the onset of advancements in the field of technology, space technology and its potential benefits are coming across in the limelight. Information derived from space research in special laboratories has been extremely efficacious in finding alternatives and solutions to situations that can be easily circumvented. The Space2030 initiative is set in a specific way that also considers the inability of developing countries to assess these resources and seeks methods for them to obtain them. International cooperation is a necessity in achieving optimal usage of these key assets. Keeping all the benefits in mind, it is necessary that throughout the implementation of the space resources, there should be no discrimination around the world for obtaining it as well as being held accountable for all the sensitive decisions made regarding it. Peace is one grave factor that cannot be compromised as it can hinder the achievement of sustainable goals. The main requirement to make this project highly successful is increasing the level of awareness surrounding the benefits and the peaks of triumphs each country can possess through it.